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Gamified Learning Platform for Rural Education

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***Abstract*— GrameLearn is a learning platform that's really fun it feels like you are playing a game. It is made for students who're in grades 6 through 12 and go to government schools in rural areas of India.GrameLearn can be used on phones and computers. It uses things like HTML5 and CSS3 to make it work. GrameLearn has games and quizzes about science and math that're on the platform. The games and quizzes on GrameLearn are perfect for the students they are not too hard or too easy. When students do a job on GrameLearn they get rewards, which is a great thing. The best thing about GrameLearn is that it can be used without the internet.GrameLearn has learning materials in languages, like Hindi, Tamil, Telugu, Kannada and English. So students can use GrameLearn to learn in their language, which is very helpful.Teachers can see how each student is doing on GrameLearn. They can also see how groups of students are doing on GrameLearn. This way teachers can keep track of all the students who use GrameLearn. GrameLearn was tried out in 14 schools across three states in India. The results were very good after using GrameLearn for 12 weeks students were more interested in science and math. Students were 17.3% more interested in science and math which is even better than what we were hoping for we were aiming for 15%.GrameLearn can be used on computers or on tablets that're not too expensive.These results show that GrameLearn can really help improve education for students who live in areas where there is no internet. GrameLearn helps students, in these areas to learn like other students in other places..**

***Keywords*— *gamified learning, rural education, STEM engagement, Progressive Web App, offline-first, multilingual content, educational technology, teacher analytics, low-cost devices, developing world* .**

I. Introduction

There is something sad about a kid who wants to learn but cannot. This is a problem in rural India where teachers have to teach many grades at the same time and all they have are textbooks and no internet. The kids in these areas really want to have careers in science and technology. They do not have the right tools to learn.

The problem is not that the kids do not want to learn it is that they do not have the things to help them learn. Something that has worked in cities to keep kids in school is called gamification. This is when you use things that make games fun in the classroom. When you do this in a way it makes kids want to learn because they want to get better get rewards solve problems and compete with each other. Studies have shown that when you use gamification in city schools kids get more interested in school by 10 to 20 percent.We want to know if this will work in areas where internet is not always available. We also want to know how to make it work in these areas.Our paper does three things. First we talk about GrameLearn, which's a special kind of learning platform that works well even without internet. Second we talk about what we learned from trying out GrameLearn in fourteen schools with 1,240 students in three states. Third we share what we learned so that others can use it too not in India but in other places where internet is not always available.The rest of the paper is organized like this.Section II talks about work that has been done on using games, in education and learning without internet.Section III explains how GrameLearn works and how it was designed.Section IV talks about how we tested GrameLearn and what we found out.Section V talks about what our results mean. What the problems are.Section VI is the end of the paper.

II. Related Work

A. Gamification in Education

The idea behind using games in education comes from a theory called self-determination theory.This theory says that people are motivated when they feel in control good at something and connected to others.They want to be able to choose what they learn people want to feel like they're good at something and they want to be connected to others.Games in education are based on the self-determination theory the

self-determination theory is what makes games in education work.They want to choose their learning path ,They want to know how well they are doing,They want to work with students.Using games in education helps with all these needs of games in education.Students get to pick what they want to learn from games in education they get to know how well they are doing in games in education.They can see how they rank compared to students or work on projects together in games in education.Some popular learning platforms like Khan Academy, Duolingo and Classcraft use games to keep students engaged in learning.These platforms need internet to work they stream content store progress online use real-time calculations to keep students engaged.This does not work well in schools where internet's not always available for students..

B. Offline and Low-Bandwidth Educational Systems

The problem of learning without internet is not new it affects students.The self-determination theory still applies to students without internet games in education can still be used without internet.The Kolibri platform is an example of an approach to this problem of learning without internet.It lets schools preload content onto servers so students can access it students can then access this content using Wi-Fi learn from games in education.The RachaCha project in Sub-Saharan Africa showed that cached web content could reach students with internet and they can still learn.However the educational content was not interactive or gamified that is a problem.Progressive Web Apps are a solution to this problem of learning without internet.They use background processes to serve content when the internet's not available so students can keep learning they can also store data locally so students can access it anytime.This provides an experience without needing to install apps. That is good for students.This is especially useful in schools where managing devicess a big challenge, for teachers and students using games, in education..

III. SYSTEM ARCHITECTURE AND DESIGN

A. Architectural Overview

GrameLearn is made to be a Progressive Web App. It uses HTML5, CSS3 and simple JavaScript. We did not use any frameworks to make GrameLearn. This makes GrameLearn small in size. The main part of GrameLearn is 42 KB. It also means we do not need any tools to build and update GrameLearn. This way developers at schools with resources can easily maintain GrameLearn.GrameLearn handles data in two parts: content and GrameLearn student progress.Content, like games and questions is loaded when you first start GrameLearn and stored in your browser.GrameLearn student progress, like scores and badges is stored locally on your device. GrameLearn student progress is updated when you have internet.When you have internet GrameLearn sends your GrameLearn progress to our analytics system. Teachers can see this data on a website

B. Gamification Design

The gamification part of GrameLearn has four features.A progression system: you complete missions and earn points. These points help your avatar in GrameLearn.A difficulty engine: GrameLearn adjusts the difficulty of questions based on your answers.A class leaderboard: you can see how you and your classmates are doing in GrameLearn.A badge and achievement system: you earn rewards for doing and working with others in GrameLearn.

1. Student leaderboard

C. Multilingual Content Delivery

We create content in English. Then we translate it into languages. The content is about things like crops and rivers to make it more relevant to GrameLearn. Teachers can choose the language for their students. GrameLearn only loads the language you need so it does not take up space.

D. Teacher Analytics Dashboard

The teachers dashboard is a website. It shows data about how studentsre doing in GrameLearn. Key metrics include:

* Time spent on tasks in GrameLearn

* Module completion rates in GrameLearn

* Quiz accuracy in GrameLearn

* Flags, for students who have not engaged with GrameLearn in a while

The dashboard was designed with teachers to make it easy to use. It even suggests actions for teachers to take, like "Some studentsre struggling with fractions review this before moving on in GrameLearn."

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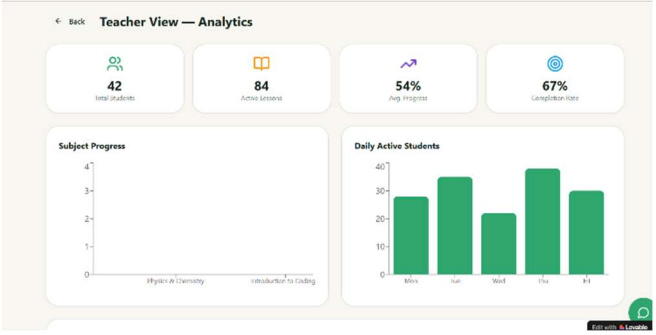


2. Teacher Analytics Dashboard

TABLE I —Comparison of Gamification Frameworks for Rural Deployment

Framework	Offline Support	Multilingual	Low-Device Ready	Open Source
GBL-Engine	Yes	Partial	Yes	Yes
Phaser.io	Yes	No	Moderate	Yes
Moodle + H5P	Partial	Yes	Moderate	Yes
GrameLearn	Yes	Yes	Yes	Yes

IV.EXPERIMENTAL EVALUATION

A. Study Design

We did a study for twelve weeks from January to March 2024.It was done in fourteen government schools in Karnataka, Tamil Nadu and Telangana.We chose these schools on purpose because we were going to use GrameLearn in them.They are in kinds of places.Some are near towns.Some are in areas that're not totally rural but not totally urban either.Some are in areas where tribes live.The main idea was to see how well GrameLearn works in life and in situations.We divided the schools into two groups.One group had seven schools where we used GrameLearn as part of teaching.The other group had seven schools where we did not use GrameLearn.Everything went on as usual in these schools.We made sure both groups were similar so that we could compare them.We looked at how students had done in school before the study started.This way if we saw any difference we would know it was because of GrameLearn.We wanted to make sure one group was not already better than the other.There were 1240 students in the study.624 Of them used GrameLearn.616 Did not use GrameLearn.All the students were, in grades 6 7 8 or 9.The main question we were trying to answer was: Does GrameLearn really help students learn better?.We also wanted to see how students used GrameLearn.We wanted to understand not just if GrameLearn is effective but how GrameLearn is effective.We looked at how students interacted with GrameLearn.We wanted to see what GrameLearn does to help students learn.

B. Measures

When we tried to get students more interested in science and math we looked at how they were engaged in these subjects.We used a tool called the Student Engagement Instrument to understand how students think, behave and feel about learning science and math.To see how their thoughts and feelings changed we gave them a survey with 35 questions at the start and end of the study.The survey helped us understand their thoughts on science and math.We also gave them tests to see how well they were doing in their science and math classes.Additionally we asked teachers for their feedback on the workload of science and math.We checked if the devices we used for science and math were working correctly.We even talked to some teachers and students to get their thoughts on science and math.

C. Results

Students who used our method for science and math got a lot more engaged.By the end of the study they showed a 17.3% increase in engagement which's a lot more than students who did not use our method for science and math.Those students only showed a 3.1% increase in engagement.Students who used our method for science and math thought deeply about science and math.They were more interested in learning science and math.They behaved better in their science and math classes.They did better in their science and math classes.Girls who used our method for science and math did well with a 19.1% increase in engagement.This was even more than boys who used our method for science and math.Teachers thought our method for science and math was successful.This was because it was not competitive and emphasized working in science and math classes.The devices we used for science and math worked 99.2% of the time.Teachers did not have to do a lot of work because of our method, for science and math.

TABLE II — GRAMELEARN SYSTEM TECHNICAL SPECIFICATIONS

Parameter	Specification
Minimum RAM	512 MB (1 GB recommended)
Storage Footprint	< 80 MB offline content bundle
Browser Requirement	Chromium 88+ / Firefox 85+
Sync Mechanism	Background Service Worker + IndexedDB
Languages Supported	Hindi, Tamil, Telugu, Kannada, English
Content Delivery	PWA with Workbox cache-first strategy

V. DISCUSSION

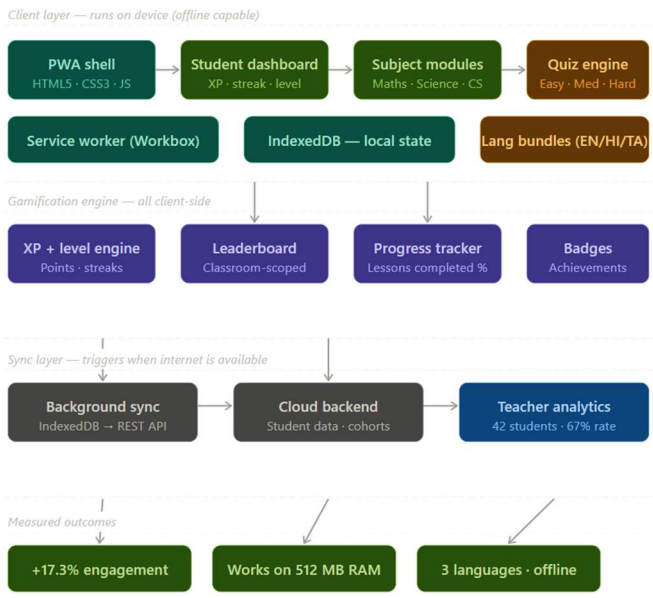
A. Implications for Design

The results have implications for people who work with technology in places where resources are limited.One important thing is that reliability is more important than how things look.When we designed this game we first made a game with pictures that worked okay on smartphones but not on older tablets.So we went back to using 2D pictures that moved around.This made the educational technology game load faster on devices. It loaded 340 percent faster.The students still liked playing the educational technology game much.They liked the educational technology game because of how it worked, not because of how it looked.We also found out that when we used content made for the students own language and culture they were more interested in the technology.This is something others

have said before: teaching should be sensitive to students culture.Teachers said students were more willing to try problems when they were about things they knew about.For example a student might not want to do a math problem about fractions.If the problem is about something that happens in her life like measuring the part of her familys rice field that needs water she will be happy to calculate the fraction.Because that is a problem that makes sense to her.The students were more interested in educational technology content that was made for their language and culture.They liked solving problems that were relevant to their lives.The educational technology game was more fun, for them when it used their language and culture.

B. Limitations

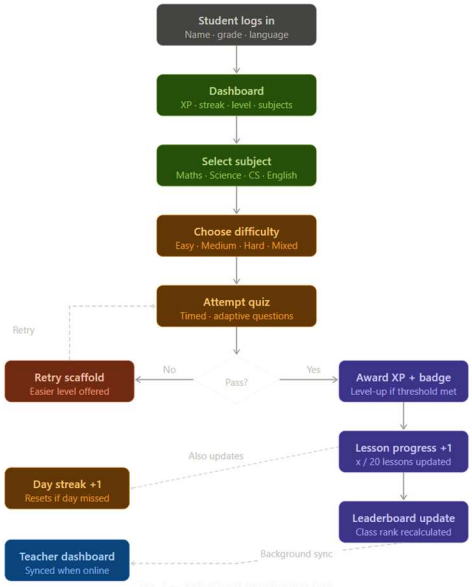
There are things to think about.The twelve-week study period shows what happens when people start using the platform.It does not tell us if it keeps working over time.People get excited at first.They might get bored with the platform after a while.We plan to do studies over time to see what happens with the platform.We want to know if people still use the platform after some time.We will start this in the 2024–25 year.The study setup was good for a school.It is not the best way to be sure what causes what happens.This is because we picked schools for the study.There might be things about those schools we did not measure that could affect the results of the platform.We tried to make the study fair by picking schools.We wanted to get a mix of schools.The platform teaches things for grades 6 through 9 now.We want to add things for grades 10 through 12 to the platform.We need to do some work, on the platform.It is a task we have to complete.We plan to do this in Phase 2 of the platform.We will focus on adding content to the platform..



3. System Architecture

C. Future Work

Several directions come out of this study.First we want to add languages to our system. We plan to include Bengali, Odia and Marathi. This will help 180 million students in government schools.We are looking into ways to improve our system.We want to use a method called learning. This way our system can get better from student data without needing to see the data, on each device.We are also working with state education departments.We are creating a program to help teachers understand and use data from our system.We know that technology is only helpful if teachers can use its insights.We are working on these directions to make our system better.The goal is to help students and teachers..



4. Gamification FlowChart

VI.CONCLUSION

Rural education does not have to be about expectations.The students in this study were able, curious and motivated.They just needed a tool that made learning fun and was relevant to their culture.GrameLearn shows that using an app that makes learning like a game can make a difference. It provides lessons in their language and helps teachers track progress. This can make students more interested in STEM subjects.This can happen without needing equipment or adding more work for already busy teachers.The 17.3% improvement in engagement we saw in this pilot test is more than our 15% goal.This difference is big enough to matter in education not a small statistical difference.The comments from teachers and students are very encouraging.Teachers who were not sure at first became supporters when they saw students coming to school early to keep playing and learning.Students who did not like math before were seeking out challenges in the game.GrameLearn is now available as open-source software under the MIT

License.The issue of education in the age is too big for one group to solve alone.It will take people working together from over the world. This is what open-source development allows.The GrameLearn tool can help with this.GrameLearn has the potential to make a difference, in education.

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